

NovaMart Intelligence Platform

Comprehensive Interface Frame Design, Component Hierarchy & Role-Based UI Architecture

UI/UX Design System Tokens & Color Palette

Designed with a state-of-the-art dark glassmorphism aesthetic (#0B0F17 base), vibrant signal accents, curated typography (Segoe UI / Consolas), and server-side RBAC component visibility.

#0B0F17 Canvas Base	#121824 Card Surface	#1E293B Border Stroke	#A100FF Accent Purple	#10B981 Emerald Success	#F59E0B Amber Alert	#F43F5E Rose Danger	#06B6D4 Cyan Telemetry
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Executive & Analytical Views

- Overview Dashboard:** KPI Cards, Priority Badges, LMDI Factor Bar Chart, 5-Signal Confidence Meter.
- Insight Analysis:** Persona Narrative, Output Validation Badge, What-If Action Sandbox, Evidence Lineage.

Governance & Diagnostics

- Abstention Protocol:** Low confidence alert, root cause failure logs, interactive clarification chips.
- Driver Attribution:** OLS Beta Table, Pearson vs Spearman matrix, Hierarchical drill-down breadcrumbs.

Interactive Labs & Admin

- Scenario Simulator:** Parameter shock sliders, sprint duration trigger, live SQLite row counters.
- Data Sources & Health:** Pipeline freshness matrix, JSON custom data ingestion, latency P95 charts.

1. Overview Dashboard UI Frame

Prioritized KPI Movements, LMDI Revenue Waterfall, Materiality Ledger & 5-Signal Confidence Gauge

FRAME 01 • OVERVIEW

Overview Dashboard Frame

Core UI Components & Interactions

- **Scenario & Role Selectors:** Dynamic dropdowns switching data payloads and server-side RBAC redaction.
- **9-Stage Pipeline Animation:** Animated modal rendering fetch → reconcile → detection → validation.
- **Materiality Ledger:** Ranks KPI shifts via score $\$ = 0.4 \text{ ext}\{\text{impact}\} + 0.3 \text{ ext}\{\text{deviation}\} + 0.2 \text{ ext}\{\text{abnormality}\} + 0.1 \text{ ext}\{\text{strategic}\} \$$.

Role-Based View Variations

- **Executive:** Full visibility across Revenue, Orders, AOV, Margin, Conversion + Action approval.
- **Marketing Manager:** Gross Margin card replaced with server-side restricted lock badge.
- **Operations Manager:** Marketing metrics stripped at API layer; inventory approvals enabled.

2. Insight Analysis & What-If Sandbox Frame

Persona Synthesis, Output Validation Badge, Interactive Decision Sliders & Auditable Evidence Ledger

FRAME 02 • INSIGHT

Insight Analysis Frame

Interactive What-If Decision Sandbox

- **Budget & Pricing Sliders:** Test counterfactual policy adjustments in real time (-30% to +30%).
- **Projected Monthly Delta:** Dynamically computes net dollar impact via regression coefficients.
- **One-Click Brief Copy:** Formats executive markdown briefing directly to user clipboard.

Guardrail Badges & Lineage Ledger

- **Output Validation Checkmark:** Confirms zero LLM hallucinations via numeric whitelist verification.
- **Associational Guard:** Rejects ungrounded causal claims; displays explicit statistical disclaimers.
- **Traceable Evidence Ledger:** Binds every narrative sentence to an evidence ID [E101-E110].

3. Selective Abstention Protocol Frame

Honest AI Refusal, Low-Confidence Warning Banner, Data Failure Logs & Clarification Chips

FRAME 03 • ABSTENTION

Abstention Protocol Frame

The 3 Machine-Enforced Abstention Triggers

- **1. Sparse History (<30 Days):** Refuses baseline analysis for newly launched products or categories.
- **2. Contradictory Evidence:** Stale marketing data (>26h) + cross-source agreement <0.50 triggers pause.
- **3. Low Confidence (<60%):** Overall 5-factor composite score falls below recommendation threshold.

Interactive Clarification Query Chips

- **Zero-Guessing Philosophy:** Replaces hallucinated advice with explicit failure diagnostic logs.
- **Interactive Chips:** Clickable prompt pills auto-populate the Analyst Chat drawer.
- **Human-in-the-Loop Recovery:** Prompts business users to refresh stale pipelines or adjust baselines.

4. Driver Attribution & Diagnostics Frame

OLS Panel Regression, Pearson/Spearman Matrix, Business Rules & IsolationForest Anomalies

FRAME 04 • DRIVERS

Driver Attribution Frame

Statistical Attribution & Correlation Check

- **OLS Panel Regression:** Displays standardized β strength bars, R^2 fit, and p-value significance.
- **Dual Correlation Matrix:** Evaluates linear Pearson r alongside rank Spearman ρ to flag non-linear drift.
- **Feedback Weight Tuning:** Analyst corrections automatically penalize disproven driver weights.

Anomaly Detection & Breadcrumb Drill-Down

- **IsolationForest Anomalies:** Unsupervised ML flags multi-dimensional outlier dates.
- **Auditable Business Rules:** Independent IF-THEN triggers (e.g. `stockout_rate > 10%`).
- **Hierarchical Drill-Down:** Interactive navigation: Revenue → Region → Category → Product.

5. Scenario Simulator & Data Sources Ingestion Frame

Interactive Parameter Shocks, Sprint Progression, Pipeline SLA Audit & Custom JSON Ingest

FRAME 05 • SIMULATOR & SOURCES

Simulator & Sources Frame

Time-Machine Simulator Controls

- **Parameter Shocks:** Marketing Spend (0.1x-3.0x), Stockouts (0-50%), Competitor Pricing (0.7x-1.3x).
- **Sprint Duration Selector:** Advance scenario time by 1, 7, or 14 days into the live database.
- **Live Row Counters:** Monitors active SQLite record additions for transactions, marketing, external context.

Data Pipeline SLA & JSON Ingest Portal

- **Freshness SLA Monitoring:** Tracks refresh latency against SLA bounds; flags stale feeds.
- **One-Click Seed:** Re-populates 365 days of reproducible mulberry32 PRNG benchmark datasets.
- **Custom JSON Ingestion:** Ingest custom order, campaign, or regional macro JSON records into SQLite.